

IWT (Internet and Web Technology) - Questions and Answers

1. Explain the architecture of the World Wide Web (WWW) with neat diagram.

Definition:

The World Wide Web (WWW) is a system of interlinked web pages and resources that are accessed through the internet using web browsers.

Architecture of WWW:

WWW mainly works on a client-server architecture. The client sends a request to the web server, and the server responds with the requested web page.

Main Components:

1. Client (Web Browser)
2. Web Server
3. Internet Connection
4. HTTP/HTTPS Protocol
5. Web Pages and URLs

Working Process:

- User enters a URL in the browser.
- Browser sends HTTP request to server.
- Server processes the request.
- Server sends response back.
- Browser displays webpage to the user.

Advantages:

- Easy information sharing
- Global communication
- Supports multimedia content
- Fast access to data

Disadvantages:

- Security risks
- Fake information
- Internet dependency

Conclusion:

WWW is one of the most important technologies that made internet communication easy and accessible worldwide.

2. What is HTML? Explain different HTML tags with examples.

Definition:

HTML stands for HyperText Markup Language. It is the standard language used to create web pages.

Features of HTML:

- Simple and easy to learn
- Platform independent
- Supports multimedia
- Used with CSS and JavaScript

Common HTML Tags:

1. Heading Tag:

```
<h1>Welcome</h1>
```

2. Paragraph Tag:

```
<p>This is a paragraph.</p>
```

3. Anchor Tag:

```
<a href="https://example.com">Visit</a>
```

4. Image Tag:

```

```

5. List Tag:

```
<ul>
```

```
<li>Item 1</li>
```

```
</ul>
```

Advantages:

- Easy webpage creation
- Supported by all browsers
- Free to use

Disadvantages:

- Static in nature
- Needs CSS for styling
- Needs JavaScript for dynamic functions

Conclusion:

HTML is the foundation of web development and is essential for creating modern websites.

3. Differentiate between GET and POST methods in web technology.

Definition:

GET and POST are HTTP methods used to send data between client and server.

GET Method:

- Data is sent through URL.
- Limited data transfer.
- Less secure.
- Used for searching and fetching data.

POST Method:

- Data is sent in request body.
- Large amount of data can be transferred.
- More secure than GET.
- Used for login forms and sensitive data.

Comparison Table:

GET Method	POST Method
Data sent in URL	Data sent in request body
Less secure	More secure

Limited data size	Large data size
Used for fetching data	Used for sensitive data

4. Explain the working of CSS and its different types with examples.

Definition:

CSS stands for Cascading Style Sheets. It is used to style HTML pages.

Working of CSS:

CSS controls colors, fonts, layouts, spacing, and animations of web pages.

Types of CSS:

1. Inline CSS:

```
<p style="color:red;">Hello</p>
```

2. Internal CSS:

```
<style>
body { background-color: lightblue; }
</style>
```

3. External CSS:

```
<link rel="stylesheet" href="style.css">
```

Advantages:

- Better design control
- Reusable styling
- Faster webpage management

Disadvantages:

- Browser compatibility issues
- Complex for beginners

Conclusion:

CSS makes websites attractive and responsive for users.

5. What is JavaScript? Explain its features and applications in web development.

Definition:

JavaScript is a scripting language used to make web pages interactive and dynamic.

Features of JavaScript:

- Lightweight language
- Object-oriented
- Event-driven
- Supports validation and animations

Applications:

- Form validation
- Dynamic content updates
- Games and animations
- Web applications
- Server-side development using Node.js

Advantages:

- Fast execution
- Improves user interaction
- Supported by all browsers

Disadvantages:

- Security risks
- Browser inconsistencies

Conclusion:

JavaScript is one of the most important programming languages for modern web development.